Project Design Phase
Solution Architecture

Date	**15 February 2026**
Team ID	**LTVIP2026TMIDS36203**
Project Name	**Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization**
Maximum Marks	**4 Marks**

Solution Architecture

The solution architecture bridges the gap between fragmented retail sales data and a centralized, interactive analytics system. It ensures structured data processing, real-time visualization, scalability, and seamless integration with business intelligence workflows.

The architecture is divided into four main layers:

1. Presentation Layer (Frontend – Visualization Interface)

Users (retail managers and analysts) interact with an interactive Tableau Dashboard embedded within a web-based interface.

- **Displays KPIs such as Total Sales, Revenue Growth, Product Performance.**
- **Provides filters (Region, Category, Time Period, Product Type).**
- **Enables drill-down analysis and dynamic visual exploration.**
- **Offers user-friendly charts, graphs, and comparative visuals.**

This layer ensures clarity, accessibility, and actionable insight delivery.

2. Logic Layer (Business & Analytical Logic)

This layer manages data transformation and analytical calculations.

- **Data cleaning and preprocessing.**
- **Calculated fields (Profit Margin, Growth %, Category Contribution).**
- **Filtering logic and parameter-based analysis.**
- **Trend analysis and comparative metrics.**
- **Conditional formatting for highlighting performance indicators.**

It ensures accurate analysis and meaningful interpretation of raw sales data.

3. Data Layer (Data Management & Storage)

This layer handles structured data storage and management.

- **Consolidates sales data from multiple sources (CSV, Excel, databases).**
- **Ensures data consistency and validation.**
- **Maintains historical records for trend comparison.**
- **Supports secure data access and retrieval.**

It enables reliable, centralized, and scalable data management.

4. Deployment Layer (Integration & Scalability)

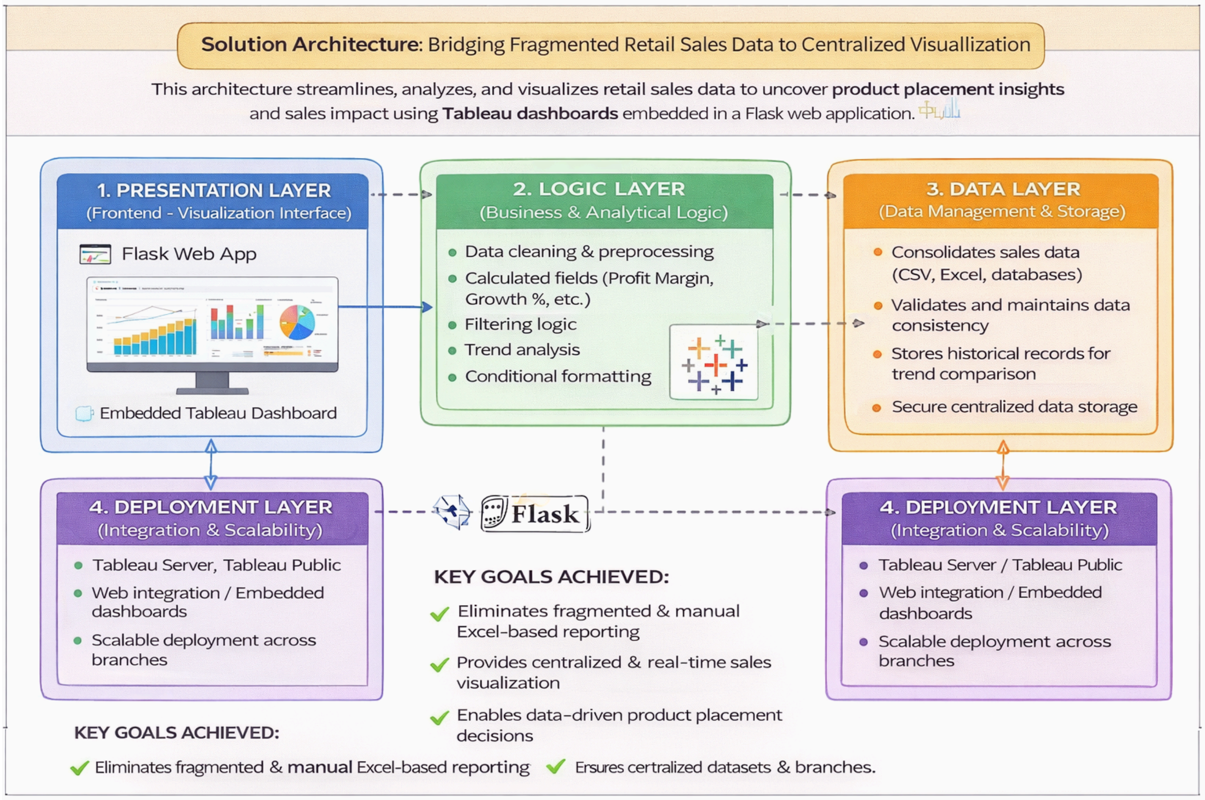
The Tableau dashboard is deployed through:

- **Tableau Server / Tableau Public / Web integration.**
- **Embedded dashboards within organizational systems.**
- **Scalable deployment across multiple retail branches.**
- **Capability to integrate additional datasets (inventory, demographics, seasonal data).**

This ensures portability, maintainability, and adaptability.

Key Goals Achieved in the Architecture

- **Eliminates fragmented and manual Excel-based reporting.**
- **Provides centralized and real-time sales visualization.**
- **Enables data-driven product placement decisions.**
- **Ensures scalability for expanding datasets and branches.**
- **Supports stakeholders with a clear understanding of data flow and analytics structure.**


Solution Architecture: Bridging Fragmented Retail Sales Data to Centralized Visualization
This architecture streamlines, analyzes, and visualizes retail sales data to uncover product placement insights and sales impact using Tableau dashboards embedded in a Flask web application.
1. PRESENTATION LAYER
(Frontend - Visualization Interface)
Flask Web App
Embedded Tableau Dashboard
2. LOGIC LAYER
(Business & Analytical Logic)
Data cleaning & preprocessing
Calculated fields (Profit Margin, Growth %, etc.)
Filtering logic
Trend analysis
Conditional formatting
3. DATA LAYER
(Data Management & Storage)
Consolidates sales data (CSV, Excel, databases)
Validates and maintains data consistency
Stores historical records for trend comparison
Secure centralized data storage
4. DEPLOYMENT LAYER
(Integration & Scalability)
Tableau Server, Tableau Public
Web integration / Embedded dashboards
Scalable deployment across branches
Flask
KEY GOALS ACHIEVED:
Eliminates fragmented & manual Excel-based reporting
Provides centralized & real-time sales visualization
Enables data-driven product placement decisions
4. DEPLOYMENT LAYER
(Integration & Scalability)
Tableau Server / Tableau Public
Web integration / Embedded dashboards
Scalable deployment across branches
KEY GOALS ACHIEVED:
Eliminates fragmented & manual Excel-based reporting
Ersures certralized datasets & branches.